1. Scope & Planning

STEP 1 - Scope & Planning | Sterile Barrier Packaging Seal - Tyvek-Film Pouch (Class III Implants)

Field	Detail
Process Name	Sterile Barrier Packaging Seal - Tyvek-Film Pouch (Class III Implants)
Scope Start	Pouching / Loading of Sterilized Implant
Scope End	Sealing, Inspection, Labeling & Distribution Release
Device Classification	Class III - PMA support process (21 CFR 814)
Standard Applied	AIAG-VDA FMEA 4th Edition 2019
Regulatory Framework	21 CFR 820 (QSR) | ISO 13485:2016 | ISO 11607-1:2019
Device Description	Sterile Tyvek-Film Pouch - Primary Sterile Barrier for Class III Implants
Team Members	Packaging Engineer, Quality Engineer, Sterilization Specialist, Supplier Quality
FMEA Facilitator	[Your Name] - Quality Engineer
FMEA Number	PFMEA-PKG-001
Revision	Rev A
Date	2026-06-22
Next Review Date	Annual review per SOP-QE-022 or upon process change
Status	Draft - Internal Review

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2. Structure Analysis

STEP 2 - Structure Analysis | System Hierarchy Breakdown

System Level	Element Name	Function Contribution
Level 1 - System	Sterile Packaging Line	End-to-end sterile barrier formation and validation
Level 2 - Subsystem	Rotary Heat Sealer	Forms hermetic seal on three open edges of Tyvek-film pouch
Level 3 - Component	Sealing Jaw / Platen	Applies controlled heat and pressure to create seal bond
Level 3 - Component	Temperature Controller (PID)	Maintains seal jaw within validated temperature window
Level 3 - Component	Dwell Timer	Controls contact time for complete seal formation
Level 3 - Component	Pouch Stock (Tyvek + Film)	Provides sterile barrier material with validated bond surface
Level 3 - Component	Operator / Packaging Technician	Loads implant, positions pouch, initiates seal cycle
Level 4 - Interface	Seal Integrity Test Station	Bubble emission (ASTM F2096) and peel strength verification
Level 4 - Interface	Label Applicator / eDHR	Applies UDI label; records lot, operator, and seal params

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3. Function Analysis

STEP 3 - Function Analysis | Intended Functions & Requirements

Process Element	Intended Function	Requirement Satisfied
Rotary Heat Sealer	Form a continuous, channel-free hermetic seal on three sides; seal width >= 6 mm per ISO 11607-1	ISO 11607-1 Annex A: seal width and channel-defect acceptance criteria
Temperature Controller (PID)	Maintain sealing jaw at 175 +/-5 C (validated window per OQ); alarm on deviation > 3 C	Seal OQ temperature mapping; process validation PQ-PKG-SEAL-001
Dwell Timer	Hold seal pressure for 1.2 +/-0.1 s (validated); prevent under-dwell causing weak seal	Seal strength OQ data; minimum peel force >= 1.5 N/15 mm per ASTM F88
Pouch Stock (Tyvek + Film)	Maintain microbial barrier and seal-ability; stored per supplier IFU (humidity/temp controlled)	ISO 11607-1 material qualification; supplier CoA per receiving SOP
Operator / Packaging Tech	Load single implant per pouch, orient correctly, position seal edge within fixture guide	Work instruction WI-PKG-001; operator qualification and annual re-certification
Seal Integrity Test Station	Detect seal channels >1 mm via bubble emission (ASTM F2096); measure peel strength per ASTM F88	ISO 11607-1 Annex A2.2; AQL sampling plan SP-PKG-02
Label Applicator / eDHR	Apply correct UDI label with lot/SN; capture seal temp, dwell, operator ID in DHR	21 CFR 830 UDI; 21 CFR 820.184 DHR; ISO 13485 7.5.3 traceability

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4. Failure Analysis

STEP 4 - Failure Analysis | Failure Modes, Effects & Causes

Process Step	Failure Mode	Effect on Patient / Downstream Process	Severity (1-10)	Root Cause	Current Prevention Control	Current Detection Control
Heat Sealing	Incomplete / channel seal defect (> 1 mm channel)	Loss of sterile barrier; microbial ingress; patient infection from non-sterile implant	9	Jaw misalignment; pouch wrinkle in seal zone; temperature excursion; worn platen	Seal jaw alignment verification at setup; temperature setpoint validated per OQ	Visual inspection per WI-VIS-004 - channels < 1 mm not reliably detected visually (GAP)
Seal Integrity	Pinhole or micro-leak in film layer (microbial ingress path)	Sterility breach; patient infection; implant recall and field safety corrective action	10	Film puncture from implant edge; handling damage; raw material defect	Pouch edge guard; gentle handling SOP; incoming film inspection	Bubble emission test samples only (ASTM F2096) - not 100% per lot (DETECTION GAP)
Temperature Control	Seal temperature out of validated range (< 170 C or > 180 C)	Under-seal: weak bond, field delamination risk. Over-seal: Tyvek fiber damage, barrier compromise	8	PID controller drift; thermocouple degradation; power supply fluctuation	PID calibration per PM schedule; operator pre-shift temperature verification	Process monitoring via controller display - no independent thermocouple alarm (GAP)
Dwell Time	Incorrect dwell time (operator override or timer fault)	Under-dwell: weak seal, peel strength below 1.5 N/15 mm; sterile barrier at risk in distribution	8	Operator bypasses timer; controller fault; recipe not locked	Timer setpoint in validated recipe; operator training	Peel strength sampled 5 pouches/lot per ASTM F88 - sub-lot variation not captured
Labeling	UDI label / content mismatch (wrong lot or implant size on label)	Incorrect device implanted; patient harm; regulatory non-conformance (21 CFR 830)	8	Manual label selection; look-alike label stock; no barcode verification at point of use	Label control SOP; label segregation; 2-person verification	Final QA label review before release - human verification only (GAP)
Post-Seal Handling	Package breach / seal peel during EtO sterilization or distribution transit	Sterility loss at point of use; patient infection; product recall	9	Over-stacking in sterilization load; inadequate secondary packaging cushioning	Stacking height limit in sterilization SOP; secondary packaging spec	Post-EtO visual inspection (sampling only); distribution simulation testing per ASTM D4169 annual

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5. Risk Rating

AIAG-VDA 2019 Action Priority (AP) -- NOT Traditional RPN

AIAG-VDA 2019 replaces RPN (SxOxD) with Action Priority (AP: High / Medium / Low). For sterile packaging, the consequence of an undetected seal defect is a non-sterile implant reaching the patient - a catastrophic outcome regardless of defect frequency. ISO 11607 requires validated sterile barrier integrity; AP prioritizes detection gaps accordingly. Any Severity >= 9 failure with Detection >= 7 is automatically High priority - RPN arithmetic can mask this by averaging a low Occurrence score.

AP Lookup: S9-10+D7-10->H | S9-10+D4-6+O>=4->H | S7-8+D7-10+O>=4->H | All others->M or L

STEP 5 - Risk Rating | AIAG-VDA Action Priority Table

Failure Mode	S (Severity 1-10)	O (Occurrence 1-10)	D (Detection 1-10)	Action Priority	AP Rationale	Legacy RPN (Reference Only)
Incomplete / channel seal defect	9	4	6	H	S>=9 + D=4-6 + O>=4: frequent critical w/ weak detection	216
Pinhole / micro-leak in film layer	10	2	8	H	S>=9 + D>=7: undetected critical failure - patient risk	160
Seal temperature out of range	8	4	5	M	S=7-8 + D=4-6 + O>=4: moderate risk profile	160
Incorrect dwell time	8	3	3	L	S=7-8 + D<=3: detection adequate for severity level	72
UDI label / content mismatch	8	2	4	L	S=7-8 + D=4-6 + O<4: controlled risk	64
Package breach during EtO / transit	9	3	5	M	S>=9 + D=4-6 + O<4: rare critical but detection acceptable	135

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6. Optimization

STEP 6 - Optimization / Corrective Actions | Risk Reduction Plan

Initial AP	Failure Mode	Action Description	Action Owner	Target Date	Revised S	Revised O	Revised D	Revised AP
H	Incomplete / channel seal defect	Implement 100% bubble emission test per ASTM F2096 (replacing sampling). Add automated peel strength tester with SPC control chart - alert at < 1.8 N/15 mm. Update Control Plan CP-PKG-001 and sampling plan SP-PKG-02. Validate test equipment per IQ/OQ IQ-SIT-001.	Packaging Engineer + QE	2026-Q3	9	2	3	M
H	Pinhole / micro-leak in film layer	Add dye penetration test per ISO 11607-1 Annex A2.2 to every batch AQL (n=32, AQL 0.65). Implement incoming film inspection with pin-hole detector per incoming SOP. Add implant edge guard (foam insert) to pouching fixture.	Quality Engineer	2026-Q3	10	2	4	M
M	Seal temperature out of range	Install independent thermocouple with dual-channel alarm: feed-hold if jaw temp deviates > 3 C from setpoint. Validate alarm response per OQ protocol OQ-SEAL-002. Reduce PID calibration interval from 12 to 6 months.	Process Engineer	2026-Q4	8	2	3	L

Initial AP	Failure Mode	Action Description	Action Owner	Target Date	Revised S	Revised O	Revised D	Revised AP
M	Package breach during EtO / transit	Add post-EtO visual inspection of all pouches before labeling (100%, not sampling). Implement ASTM D4169 distribution simulation testing semi-annually (current: annual). Add foam separator between pouch layers in sterilization cassette.	Packaging + Sterilization PE	2026-Q4	9	2	3	M

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7. Results Summary

STEP 7 - Results & Documentation Summary

pFMEA Metrics Summary (auto-computed)

Total Failure Modes Analyzed	6
Initial High (H) Action Priority	2
High AP After Actions Implemented	0
Medium (M) AP - Initial	2
Medium (M) AP - After Actions	3
Low (L) AP - After Actions	3
Total Corrective Actions Assigned	4
Actions with Owner Assigned	4
Actions with Target Date	4

Linked Quality System Documents

Document Type	Document Number	Title / Description	Status
Control Plan	CP-PKG-001 Rev A	Sterile Packaging - Heat Seal Control Plan	In Review
Work Instruction	WI-PKG-001 Rev C	Tyvek-Film Pouch Sealing - Setup & Operation	Released
Inspection Plan	SP-PKG-02 Rev B	Seal Integrity Sampling Plan - Bubble Emission & Peel Strength	Released
OQ Protocol	OQ-SEAL-002	Heat Sealer Temperature & Dwell Time - Operational Qualification	In Progress
PQ Protocol	PQ-PKG-SEAL-001	Process Validation - Sterile Packaging Transfer	Planned
Risk Management	RM-PKG-2024-01	Packaging Risk Management File (ISO 14971)	In Review
Sterilization PQ	PQ-ETO-HIP-001	EtO Sterilization Validation - Class III Implant Family	Released
SOP	SOP-QE-022	pFMEA Creation and Maintenance Procedure	Released

Archive Location: [PLM System] -> Projects -> PKG-SEAL-TRANSFER-2024 -> FMEA -> PFMEA-PKG-001_RevA.xlsx

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